



# INTRO TO MACHINE LEARNING

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# WHAT IS MACHINE LEARNING?

- “Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed”

—Arthur L. Samuel, AI pioneer, 1959

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# WHAT IS MACHINE LEARNING?

- A program is said to learn from experience  $E$  with respect to some task  $T$  and some performance measure  $P$ , if its performance on  $T$ , as measured by  $P$ , improves with experience  $E$ .

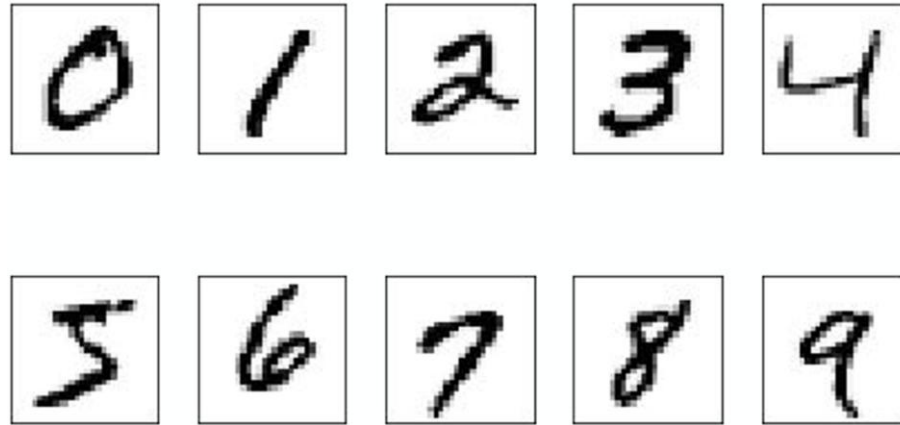
— Tom Mitchell, Professor at Carnegie Mellon

- Experience: Data
  - Task: Prediction, etc.
  - Performance: RSME, etc.
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# MACHINE LEARNING TASKS

Handwriting Recognition Example:



- Experience:
- Task:
- Performance:

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# WAIT, I'VE ALSO HEARD ABOUT STATISTICAL LEARNING...

- Statistical Learning = Machine Learning (ish)
    - “Statistical” learning emphasizes modern ML’s roots and foundation in probability and statistics for handling uncertainty
  - So what is statistical/machine learning?
    - A field of study concerned with making quantitative inferences and predictions from data
    - Uses statistical techniques and computational power to “learn” these inferences and predictions without being explicitly programmed
    - A little more formally: an algorithm is said to learn from data if its performance improves as the amount of data increases
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# SUCCESS OF MACHINE LEARNING

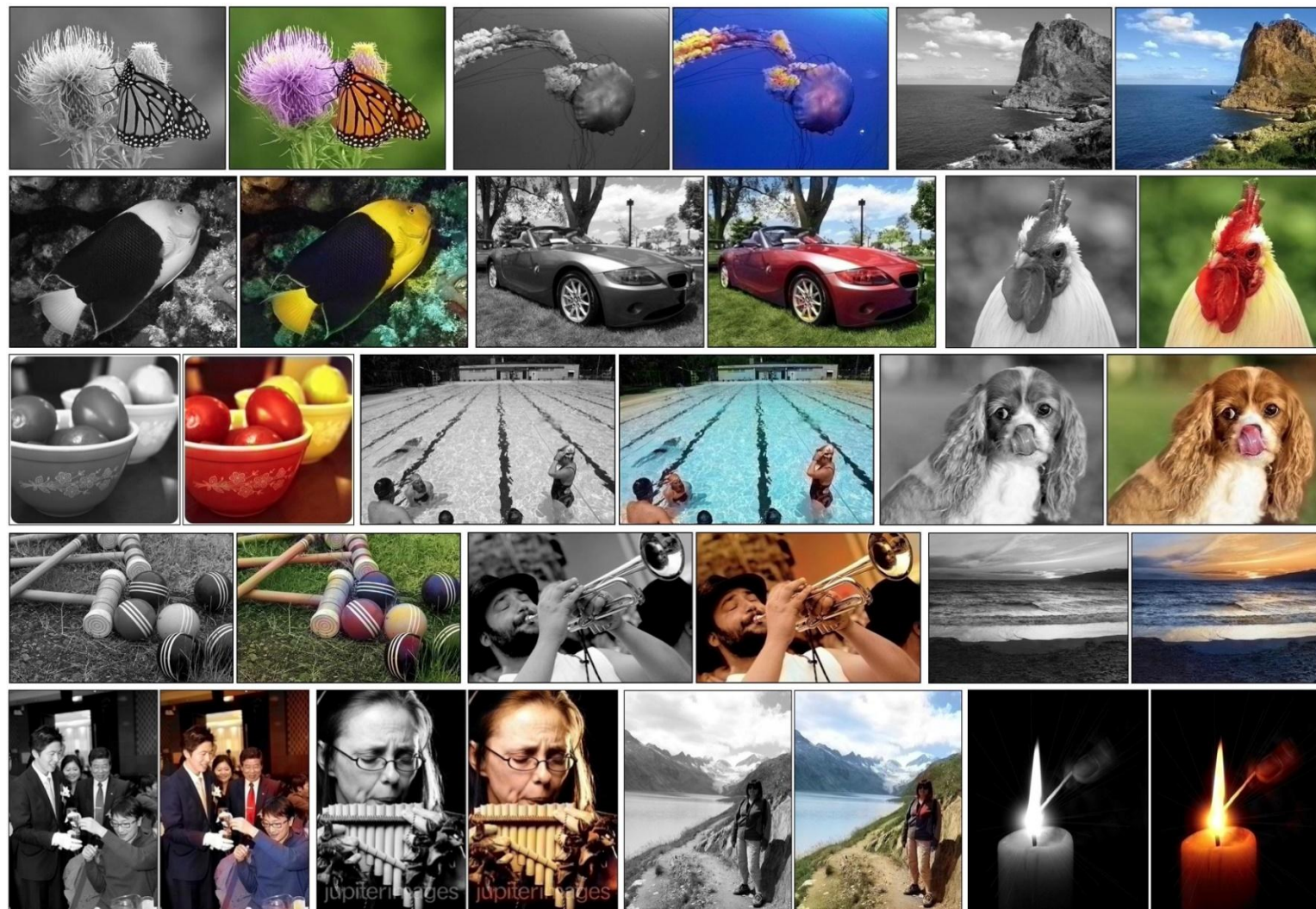
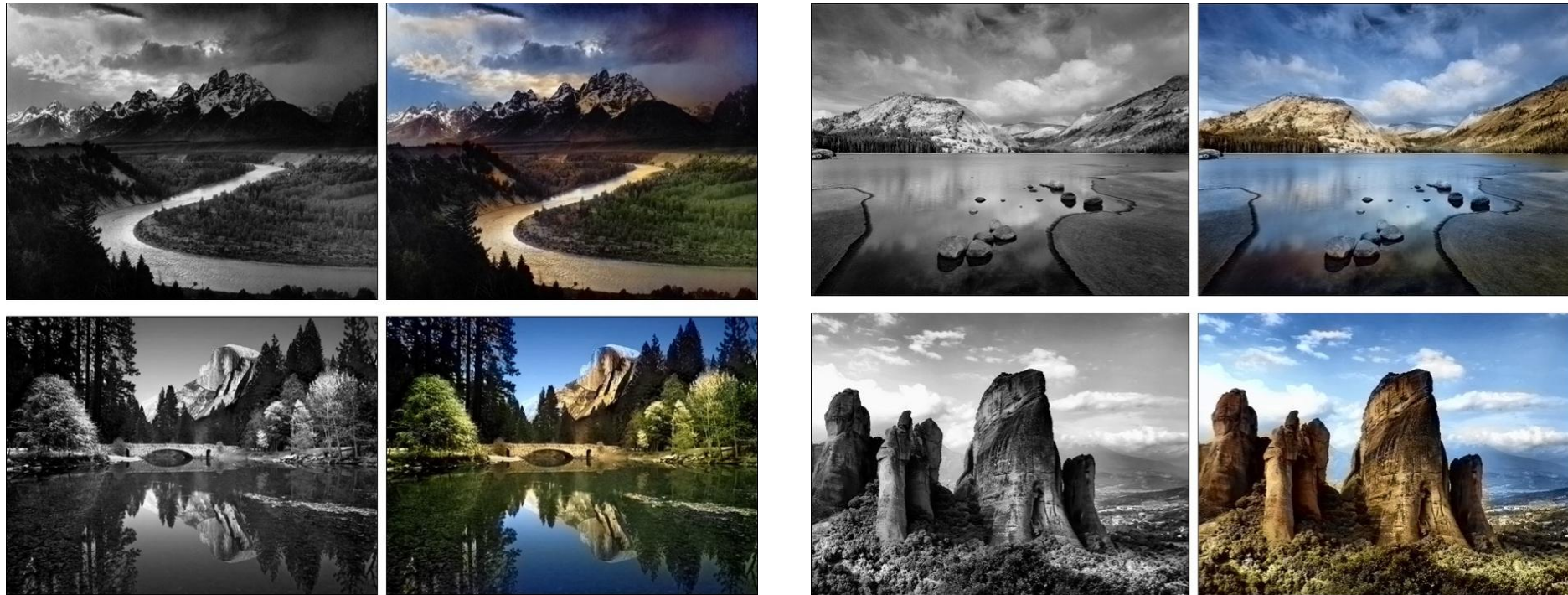


Image colorization  
(Zhang et al., 2016)

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# SUCCESS OF MACHINE LEARNING

Image colorization (Zhang et al., 2016)



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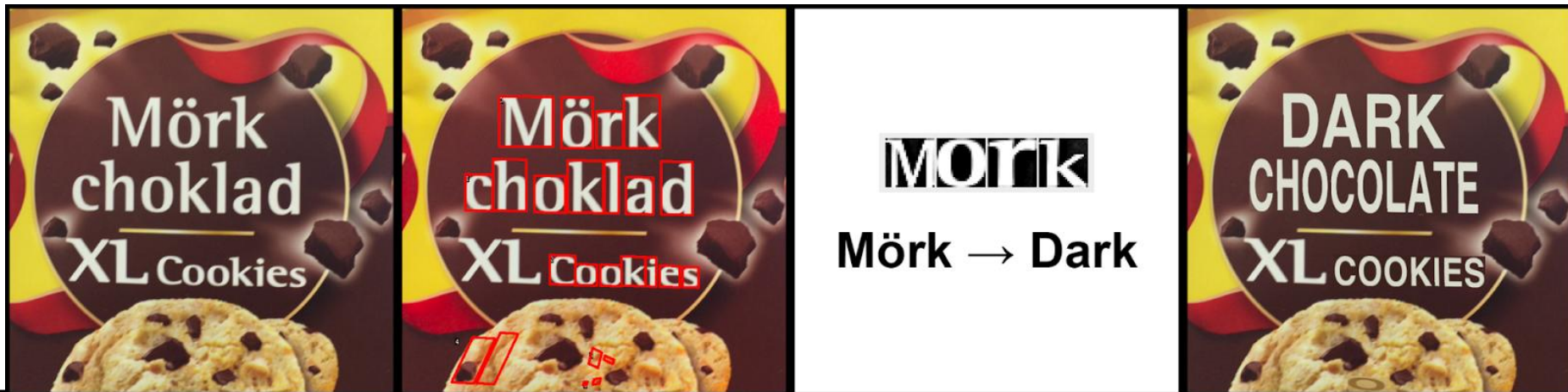
Colorized classical photographs by Ansel Adams

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# SUCCESS OF MACHINE LEARNING

Real-time visual translation on smartphones

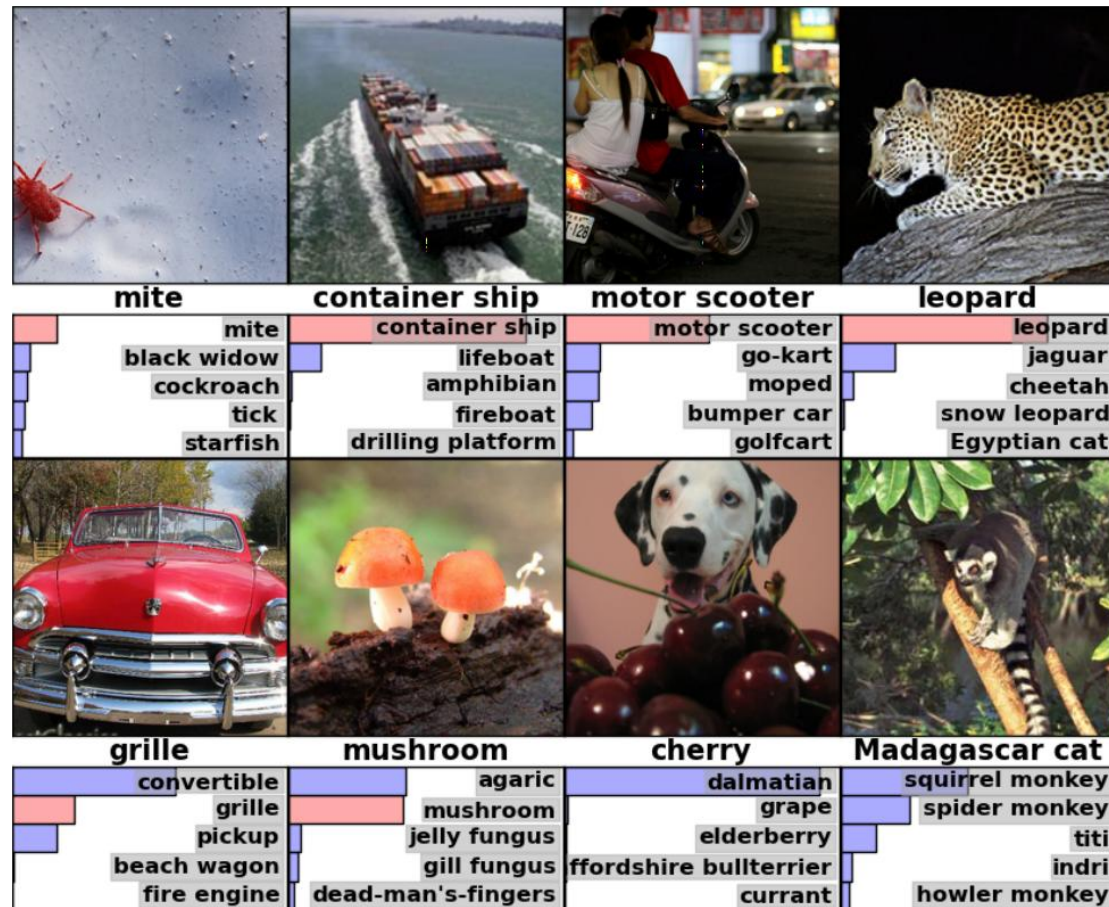
1. Find the letters
2. Recognize the letters
3. Translate
4. Render the translation in the same style



Google blog, 2015

# SUCCESS OF MACHINE LEARNING

Object classification/detection in images (Krizhevsky et al., 2012)



# SUCCESS OF MACHINE LEARNING

## Alpha Go Zero

  
International journal of science

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Altmetric: 2188 Citations: 1 [More detail >>](#)

Article

### Mastering the game of Go without human knowledge

David Silver , Julian Schrittwieser, Karen Simonyan, Ioannis Antonoglou, Aja Huang, Arthur Guez, Thomas Hubert, Lucas Baker, Matthew Lai, Adrian Bolton, Yutian Chen, Timothy Lillicrap, Fan Hui, Laurent Sifre, George van den Driessche, Thore Graepel & Demis Hassabis

*Nature* **550**, 354–359 (19 October 2017)  
doi:10.1038/nature24270  
[Download Citation](#)

Received: 07 April 2017  
Accepted: 13 September 2017  
Published online: 18 October 2017

Computational science  
Computer science Reward

## Alpha Zero

arXiv.org > cs > arXiv:1712.01815 Search or .  
(Help | Advance)

Computer Science > Artificial Intelligence

### Mastering Chess and Shogi by Self-Play with a General Reinforcement Learning Algorithm

David Silver, Thomas Hubert, Julian Schrittwieser, Ioannis Antonoglou, Matthew Lai, Arthur Guez, Marc Lanctot, Laurent Sifre, Dhharshan Kumaran, Thore Graepel, Timothy Lillicrap, Karen Simonyan, Demis Hassabis

(Submitted on 5 Dec 2017)

The game of chess is the most widely-studied domain in the history of artificial intelligence. The strongest programs are based on a combination of sophisticated search techniques, domain-specific adaptations, and handcrafted evaluation functions that have been refined by human experts over several decades. In contrast, the AlphaGo Zero program recently achieved superhuman performance in the game of Go, by tabula rasa reinforcement learning from games of self-play. In this paper, we generalise this approach into a single AlphaZero algorithm that can achieve, tabula rasa, superhuman performance in many challenging domains. Starting from random play, and given no domain knowledge except the game rules, AlphaZero achieved within 24 hours a superhuman level of play in the games of chess and shogi (Japanese chess) as well as Go, and convincingly defeated a world-champion program in each case.

Subjects: **Artificial Intelligence (cs.AI)**; Learning (cs.LG)  
Cite as: [arXiv:1712.01815 \[cs.AI\]](#)  
(or [arXiv:1712.01815v1 \[cs.AI\]](#) for this version)

#### Submission history

From: David Silver [[view email](#)]  
[v1] Tue, 5 Dec 2017 18:45:38 GMT (272kb,D)

Automatic game playing

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# SUCCESS OF MACHINE LEARNING



Find the bug with this code:

```
...  
for (var i = 0; i < 5; i++) {  
  setTimeout(() => console.log(i), 1000)  
}  
...
```



The code contains a bug that will cause it to always print the number 5 to the console, rather than the expected sequence of numbers from 0 to 4. This is because the `setTimeout` function creates a closure that captures the value of the `i` variable, but the `i` variable is changed before the callback function is executed.



To fix this bug, the code can be modified to create a new variable in the loop that is passed to the `setTimeout` callback function, rather than using the loop variable directly. This will ensure that each callback function captures the correct value of the loop variable, and the expected output will be produced. Here is an example of how the code could be fixed:

```
Copy code  
  
for (var i = 0; i < 5; i++) {  
  // Create a new variable and pass its value to setTimeout  
  const value = i;  
  setTimeout(() => console.log(value), 1000)  
}
```

This code will print the numbers 0 to 4 to the console, as expected. Alternatively, the `let` keyword can be used in place of `var` to declare the loop variable, which will automatically create a new variable for each iteration of the loop and avoid the need to create a new variable manually. This is a common pattern for avoiding closure-related bugs in JavaScript.

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# SUCCESS OF MACHINE LEARNING

AI generated art



Best Dalle2 Pics  
@Dalle2Pics

...

A raccoon playing tennis at Wimbledon in the 1990s  
[#dalle2](#) [#dalle](#)



3:33 AM · May 14, 2022

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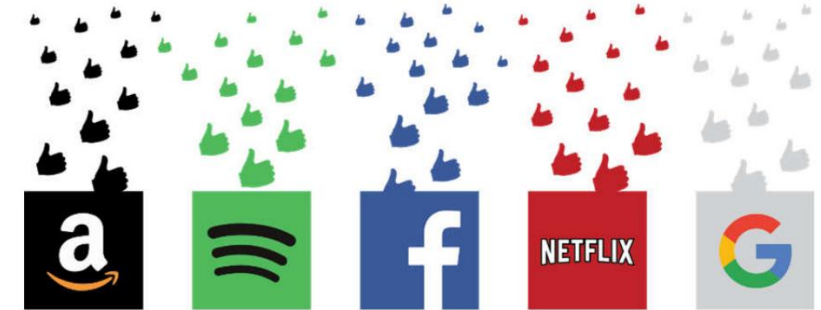
# I'M JUST CURIOUS...

- What are some of the most interesting things you have used ML/AI for?

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# OTHER MACHINE LEARNING APPLICATIONS

- Chatbots (NLP)
- Fraud detection
- Route planning
- Search engine refinement
- Face recognition/detection
- Credit risk assessment
- Financial market prediction
- Medical diagnosis
- Personalized medicine
- Electricity demand forecasting
- Spam filtering
- Collision avoidance systems
- Speech synthesis/analysis (e.g. Siri, Alexa)
- Recommender systems (e.g. Netflix, Spotify)



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# CATEGORIES OF ML

- Supervised Learning
    - Labeled Data
    - Predict Outcome
    - Direct Feedback (E.g., accuracy, RMSE. Etc.)
  - Unsupervised Learning
    - No Labels/Targets
    - “Fuzzy” Feedback: No Target = No Objective Less Clear
  - Reinforcement Learning
    - Set of states/actions
    - Maximize “reward”
-

| Term used in Statistics            | Term used in Machine Learning       |
|------------------------------------|-------------------------------------|
| Observation / case                 | <b>Example / instance</b>           |
| Explanatory variables / predictors | <b>Features / input variables</b>   |
| Response variable                  | <b>Target / label</b>               |
| Parameter estimates                | <b>Learned parameters / weights</b> |
| Intercept                          | <b>Bias term</b>                    |
| Fitted model                       | <b>Trained model</b>                |
| Fitted value                       | <b>Prediction</b>                   |

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# WHY ESTIMATE F?

- Prediction
  - Inputs  $X$  are available, but the output  $Y$  is unknown or future.
  - $\hat{Y} = \hat{f}(X)$ : a black-box model can be fine if it predicts accurately.
- Inference
  - Understand how  $Y$  is related to  $X$ .
  - Which predictors matter? What is the direction/size of each relationship?
  - Many real problems require both prediction and inference.

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# GOALS OF SUPERVISED ML

- Prediction
  - Predict the target for new observations
  - Forecast the target for future observations
- Interpretation
  - Which feature are associated with the target?
  - What is the relationship between target and each feature?
  - Are some features more important than others?
  - Is there an explainable grouping in the data?

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# SUPERVISED LEARNING

- The learner/user are given training data

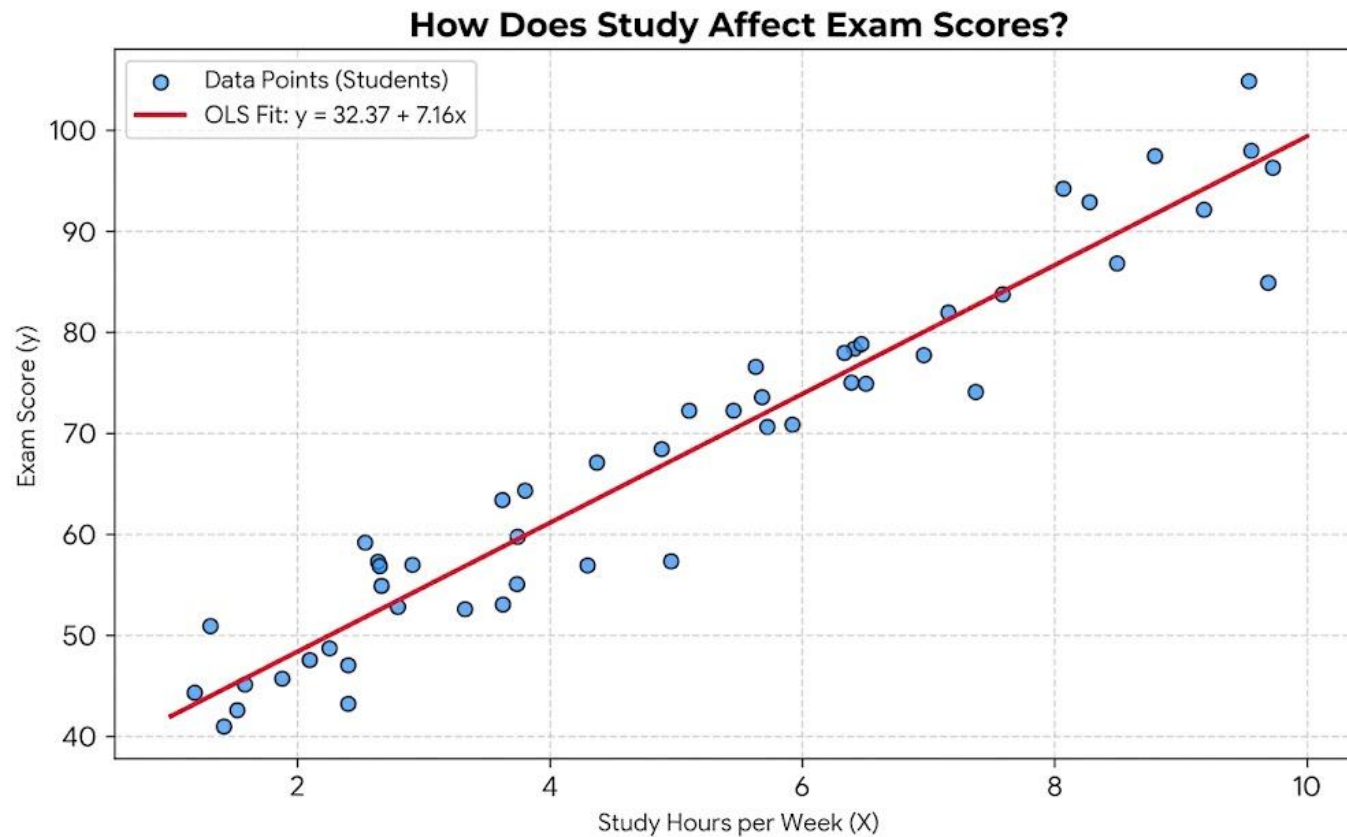
$$(x_1, y_1), \dots, (x_n, y_n)$$

- Each  $y_i$  is the output of an unknown and possibly noisy function with input  $\mathbf{x}_i$
  - Goal: Given a test input  $\mathbf{x}$ , predict the correct output
  - Two primary types of supervised learning: classification and regression
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# REGRESSION

Outputs are called responses and are real valued:  $y \in \mathbb{R}$



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# CLASSIFICATION

- Outputs are called labels and are typically finite:

$$y \in \{1, 2, \dots, C\}$$

where  $C$  is the number of classes

- Example: handwritten digit recognition



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# UNSUPERVISED LEARNING

- Data is unlabeled
  - $(x_i)$  and not  $(x_i, y_i)$
  - **Goal:** infer a property from the data, not to predict a property from future data
  - Examples:
    - **Clustering:** What are the distinct clusters or groups in a dataset?
    - **Density Estimation:** What is the probability density function that generated the data?
    - **Dimensionality Reduction:** How can we reduce the dimensionality of the data without losing (much) information?
    - **Anomaly Detection:** Automatically identifying outliers.
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# INSTANCE-BASED VS MODEL-BASED

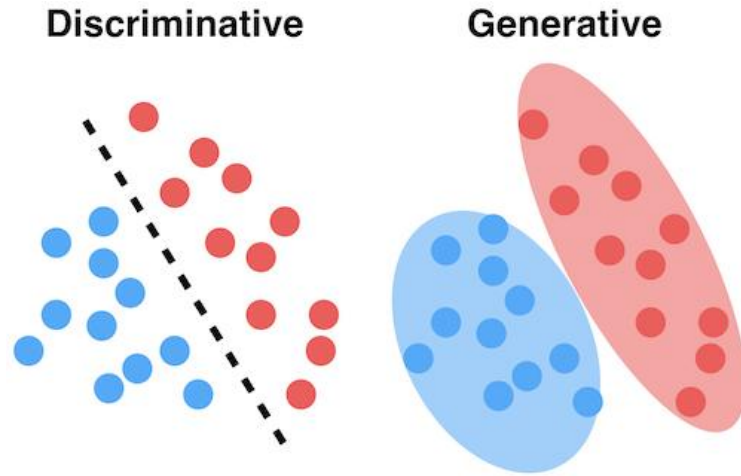
- Instance-Based
  - Learn training data by heart
  - Generalize to new observations based on *similarity* to existing observations
    - Requires a *metric* that quantifies similarity
  - Example: **k** Nearest **N**eighbors (k-NN)

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# INSTANCE-BASED VS MODEL-BASED

- Model-Based
    - Estimate the model  $f(x)$  that defines the relationship between  $X$  and  $y$
    - Predictions are made by introducing new observations into the model:  $\hat{y}_{new} = f(x_{new})$
    - Most ML algorithms are model-based (e.g., linear regression, neural networks)
    - Model-based algorithms are trained by defining an *objective function* and finding the parameters or values that *optimize* the objective function
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# SOME TERMS



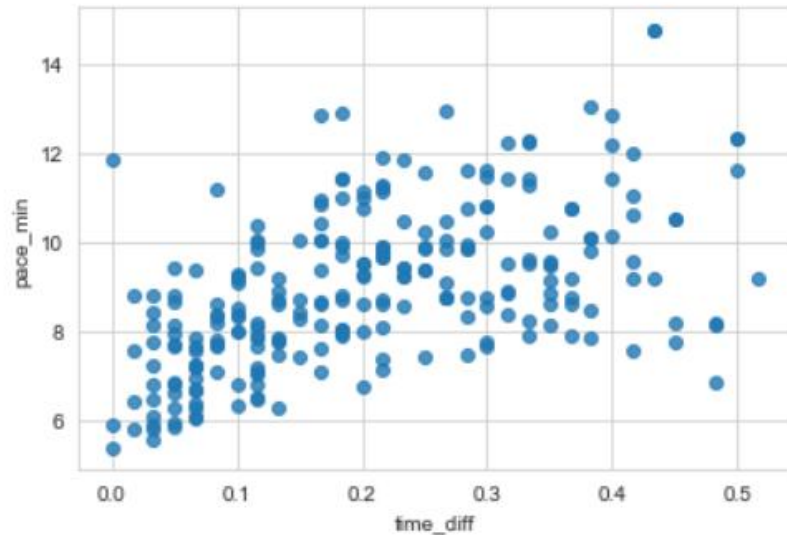
Some adjectives used to describe ML algorithms

- **Distributional assumptions**
  - Generative: assumes a full probabilistic model of the observed data
  - Discriminative: assumes a partial or no probabilistic model
- **Computational form**
  - Linear: produces a linear/affine function
  - Nonlinear: not linear
- **Complexity**
  - Parametric: # of model parameters\* is independent of sample size
  - Nonparametric: # of model parameters\* grows with sample size
- **Learning Type**
  - Lazy learner: No explicit model is built at training time (usually **instance-based**).
  - Eager learner: A general model is built during training.

\*Model parameters = parameters needed to represent the function produced by the learning algorithm

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# EXAMPLE: SIMPLE LINEAR REGRESSION



How does simple linear regression work?

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# EXAMPLE: SIMPLE LINEAR REGRESSION

- Objective function: (this is a *cost* function since we want to find the *minimum*)

$$L(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2$$

- We find the  $\widehat{\beta}_1$  and  $\widehat{\beta}_0$  that minimize the objective function
  - Optimization method:
    - *How* do we perform the optimization?
-

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# GENERAL MODEL-BASED TRAINING FRAMEWORK

- Define an appropriate **objective function**
  - Cost/Loss functions are minimized
  - Utility/Fitness functions are maximized
    - For example the likelihood in MLE is a utility function
- “Train” the model by solving the optimization problem using an appropriate optimization **method**
  - a “closed-form” solution (rare)
  - a numerical method (usually)

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# CHALLENGES IN ML

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# INTERPRETABILITY VS FLEXIBILITY

(Graphic from ISLP)



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# CHALLENGES IN MACHINE LEARNING

- Big data (both sample size and dimension)
  - Imbalanced labels
  - Noisy labels
  - Missing labels
  - Missing features
  - Uncertainty
  - Data representations
  - Safety (e.g., self-driving cars)
  - Fairness
  - Ethics
  - Explainability/Interpretability
  - Changing environment
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# THE IMPORTANCE OF DATA

- Data is the fuel that powers our learning machine
- No amount of model improvement can compensate for “bad” data
  - “Bad” data is not always poor quality
  - Sometimes “bad” data is just the wrong data for the task



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# THINGS TO ASK OF YOUR DATA

- What data is available?
  - Is the data still relevant?
  - What other data sources might be available?
  - Is there *enough* data?
  - Where did the data come from?
  - Do you know and trust the data source?
  - Does the available data actually answer the question?
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# POTENTIAL BIASES IN DATA

- **Selection bias:** Data is selected for convenience or self-selected
  - **Confirmation bias:** We are less critical of data (and results) that confirm our prior beliefs
  - **Societal bias:** Data often reflects cultural stereotypes
  - **Measurement bias:** Biases due to limitations of measurement devices
  - **Proxy measurement bias:** A proxy measurement is used for something that is hard to measure
  - **Exclusion bias:** Important variables might not be measured
  - **Sampling bias:** Data isn't representative or doesn't reflect realities of where the ML model will run
  - **Human bias:** All data is somehow collected by humans
-